PSOC™ Edge E84 training manual for Edge Protect Bootloader

About this document

Scope and purpose

This document is a training manual for the hands-on exercise with an Edge Protect Bootloader on PSOC™ Edge using ModusToolbox™.

Intended audience

The intended audiences for this document are design engineers, technicians, and developers of electronic systems.

Table of contents

1 Introduction

This manual provides specific instructions to create, configure , build, and run the code examples on the PSOC™ Edge E84. In this lab session you will learn how to use Edge Protect Bootloader on PSOC™ Edge and enabling chain of trust with EPB.

2 Required development tools

- ModusToolbox™ software v3.6 or later
    - Recommended installation via ModusToolbox™ Setup tool
- Eclipse IDE for ModusToolbox™ v2025.4.0 or later
    - Recommended installation via ModusToolbox™ Setup tool
- Edge Protect Security Suite v1.6 or later.
    - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.5.0 or later
    - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- Board support package (BSP):
    - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Terminal emulator
    - Instructions in this document use Tera Term
- PSOC™ Edge E84 Evaluation Kit (KIT_PSE84_EVAL)
    - Ensure device boots from external memory:
        - BOOT SW/SW6 in 'ON/HIGH' position
    - Disable alternate serial interface configuration:
        - J20/J21 in Tristate/Not-Connected (NC) position

3 Prerequisites

Install the software and obtain hardware listed in the Required development tools section.

This class will not cover basic concepts of ModusToolbox™ and PSOC™ Edge.

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, visit:
    - https://www.infineon.com/product-information/psocdeveloper
- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit:
    - PSOC™ Edge E84 Training Collection

4 Basic Secure App with EPB

4.1 Objective

Designed for beginners, this lab session provides a step-by-step approach to start using the Edge Protect Bootloader along with a simple application (Basic Secure App code example).

4.2 Description

This code example demonstrates the following flow:

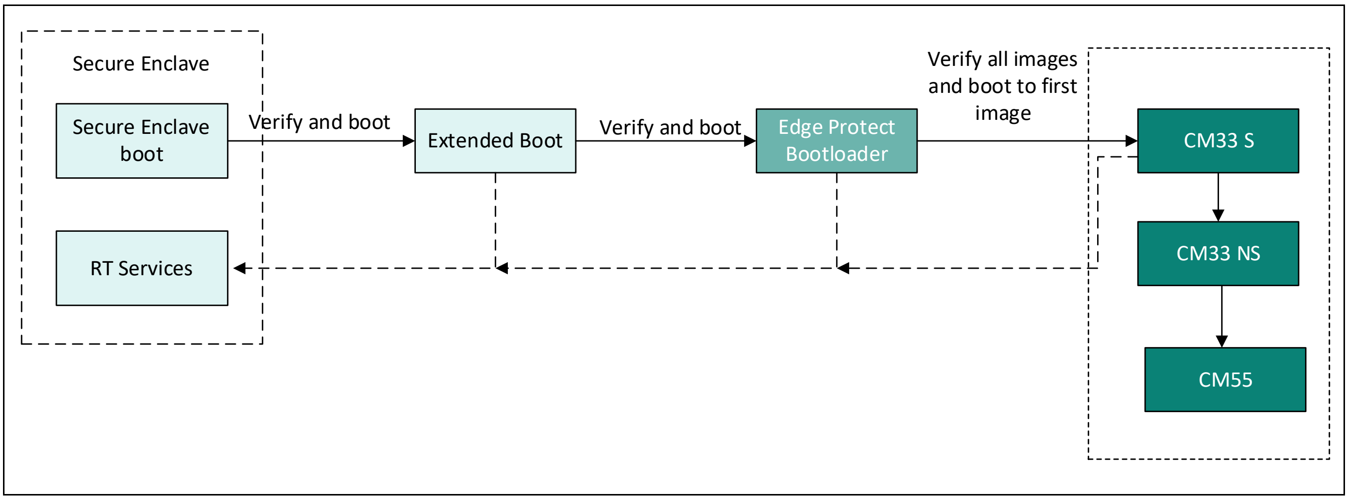


Figure 1 Boot sequence with Edge Protect Bootloader

For details of the terminology used, refer to AN237849 Getting Started with PSOC™ Edge Security.

4.3 Project Creation

1. Before creating the first project we need to create our workspace in ModusToolbox™. Open **ModusToolbox**™ from the **Windows Start** menu.

Note: If you have not installed ModusToolbox™, see the Required development tools section.

Note: If you have already created a workspace, please ignore this step and move on to step #4 directly.

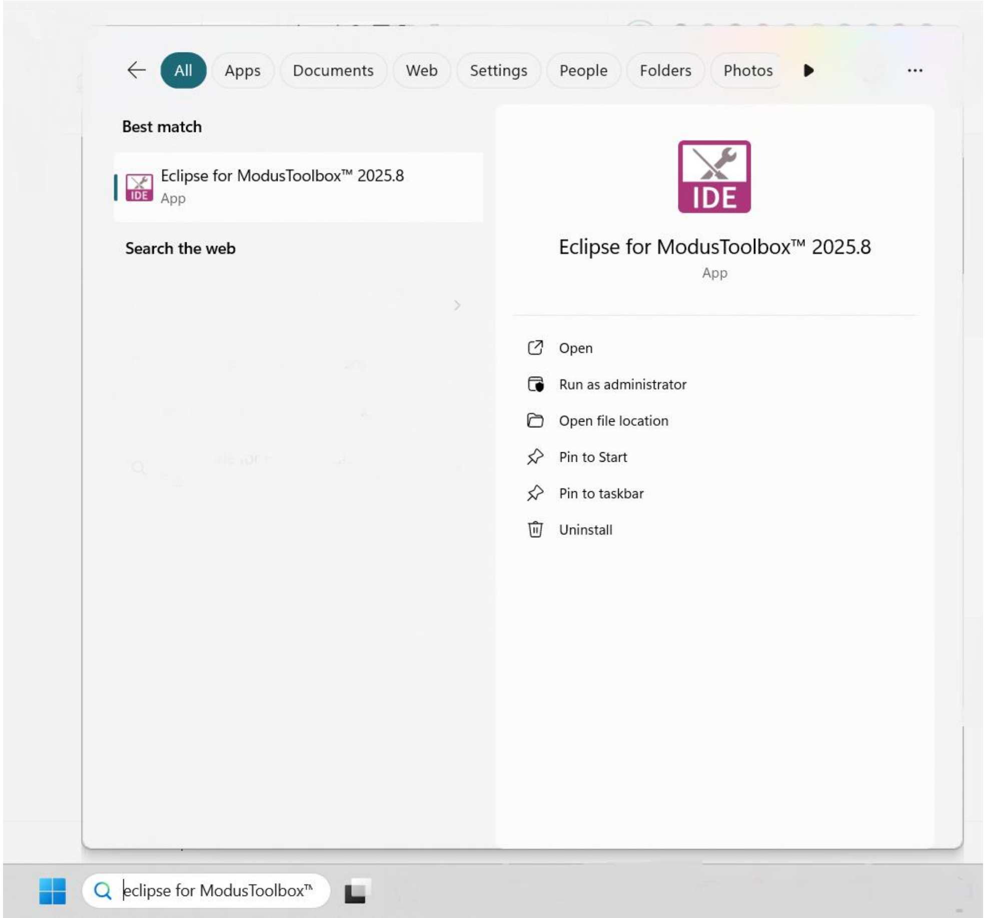


Figure 2 **Start ModusToolbox™**

2. **Next**, choose a **workspace directory** for your project. After selecting a suitable directory, click the **Launch** button.

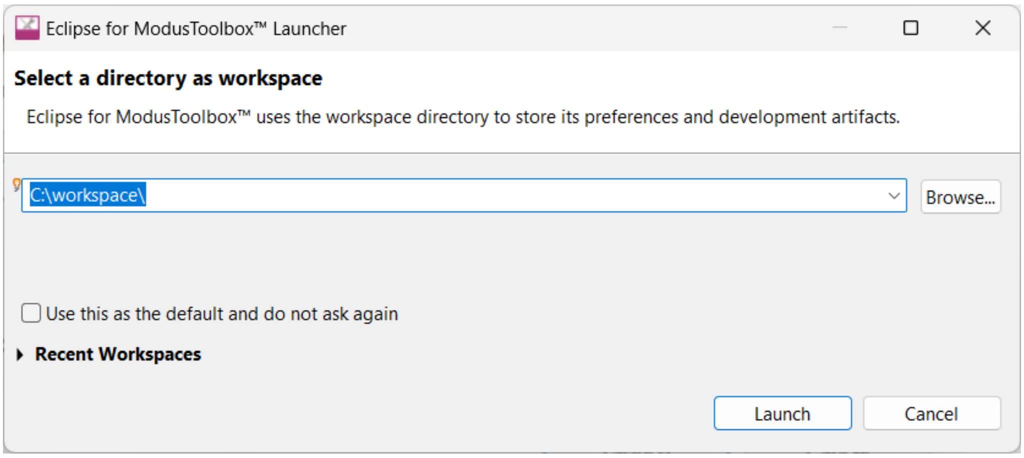


Figure 3 **Select workspace directory**

3. Clicking Launch opens a new Modus™ Toolbox workspace as shown in Figure 5.

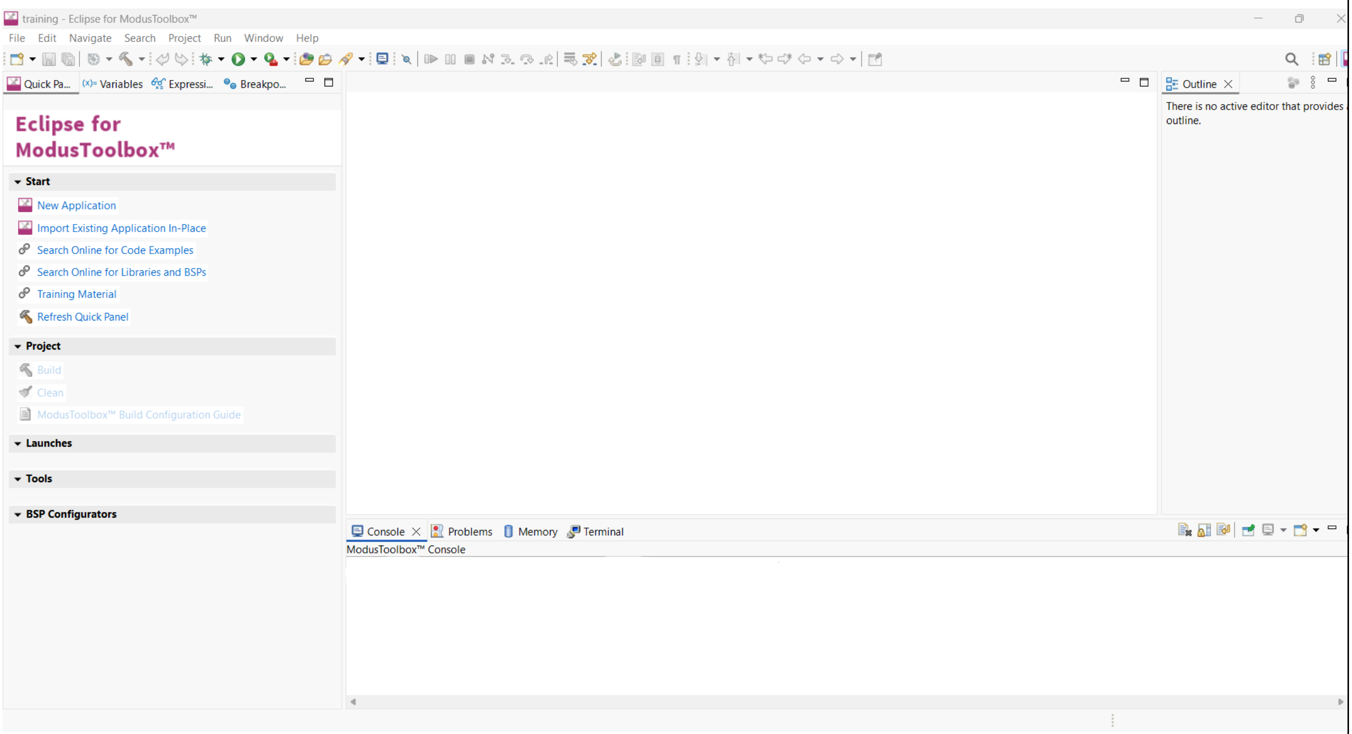


Figure 4 **Launch MTB workspace**

4. Now, select **New Application** from the Quick Panel. Alternatively, go to **File> New> Modus Toolbox™** Application.

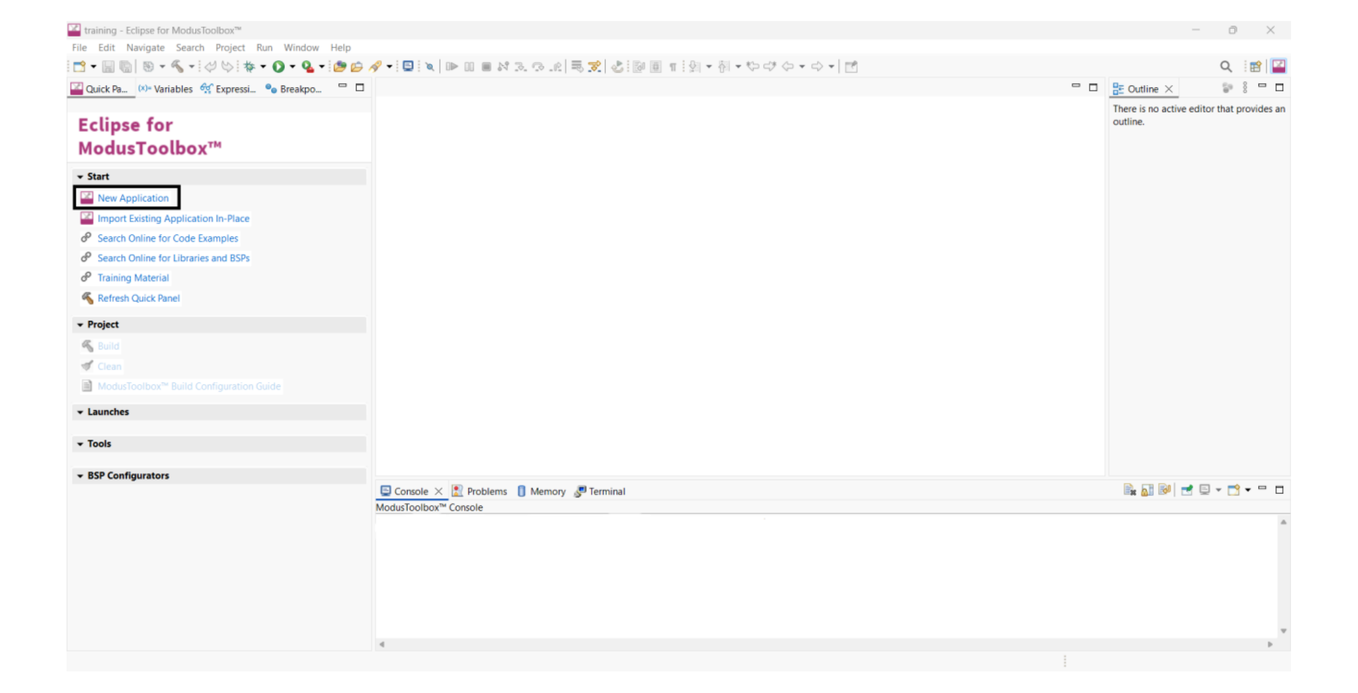


Figure 5 **Create new application**

5. Clicking New Application opens the **Project Creator Tool** and prompts you to choose a **board support package** (BSP).

*Note: BSPs are aligned with our development/evaluation kits; they provide files for basic device functionality. A BSP typically has a **design.modus** file that configures clocks and other board-specific capabilities. That file is used by the ModusToolbox™ configurators. A BSP also includes the required device support code for the device on the board. You can modify the configuration to suit your application.*

6. Once **Projector Creator** launches, Select the **KIT_PSE84_EVAL_EPC2 BSP** under the PSOC™ Edge BSPs. Click Next.

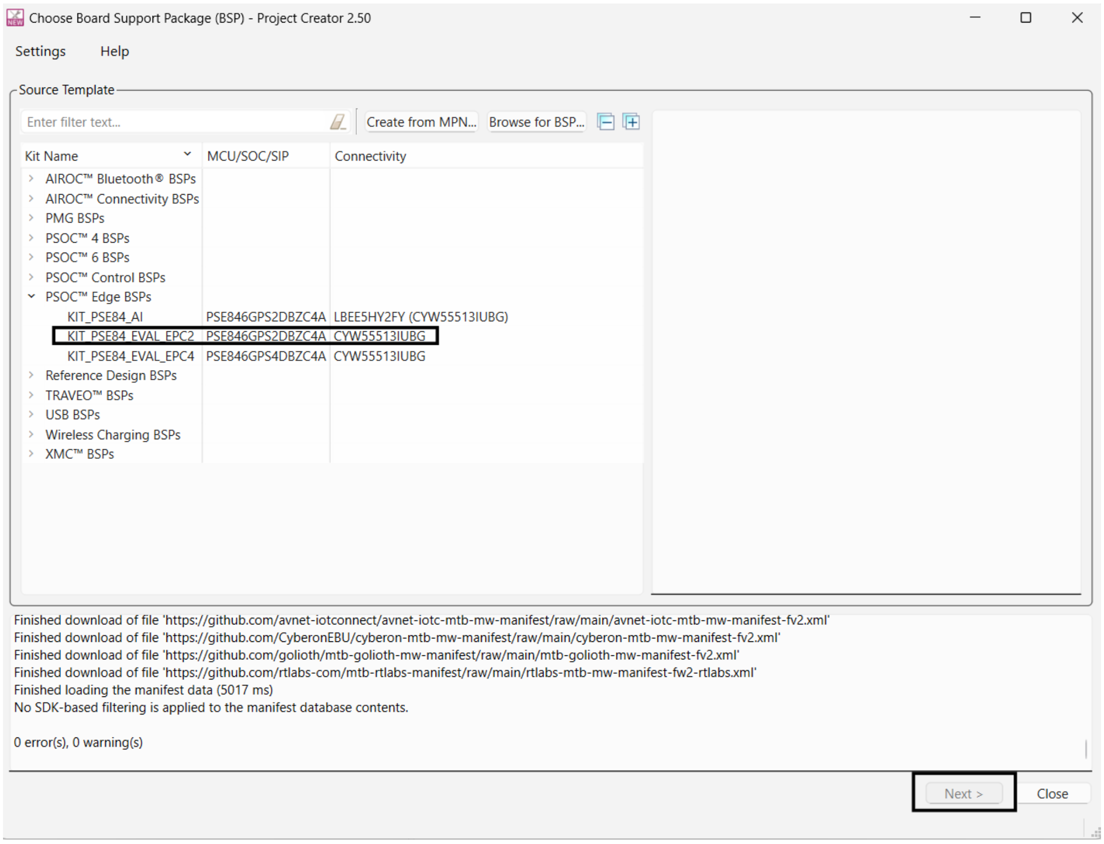


Figure 6 Select BSP in project creator

7. Clicking Next takes you to the next menu of the Project Creator which asks you to select an application.
   a) Select the **PSOC™ Edge Protect Bootloader** under the **Security** section. You can do this by selecting the **checkbox** near the application.
   b) Select the **PSOC™ Edge Basic Secure App** under the **Security** section. You can do this by selecting the **checkbox** near the application.
   c) Click **Create**.

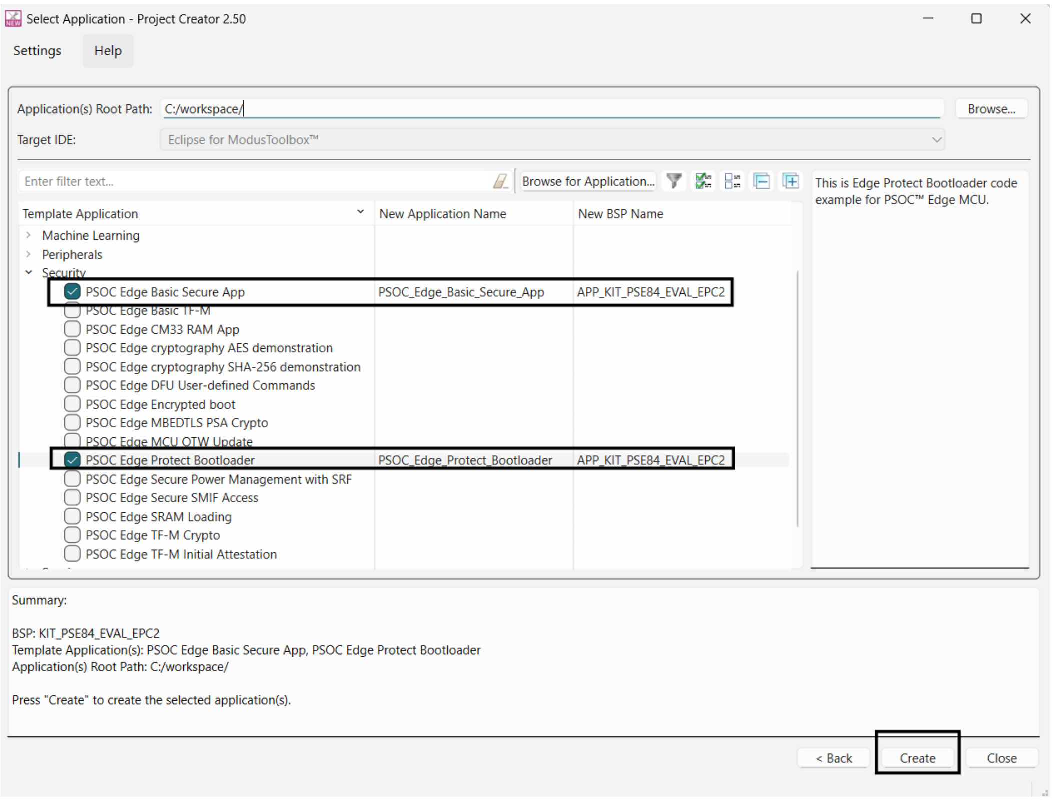


Figure 7 **Select application in project creator**

Note: *Complete demonstration of Edge Protect Bootloader requires Basic Secure App. Creating both code example together in above step.*

8. When you click **Create**, two new ModusToolbox™ applications are created. You can see both applications in Project Explorer window as shown in Figure 8.

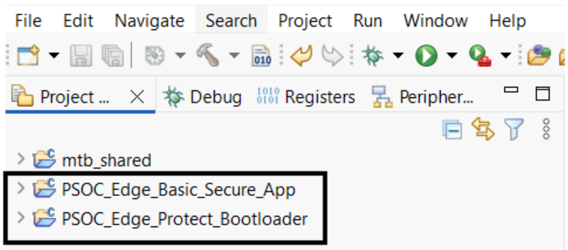


Figure 8 **Project explorer window**

9. Copy proj_bootloader under Edge Protect Bootloader by right clicking on the it.

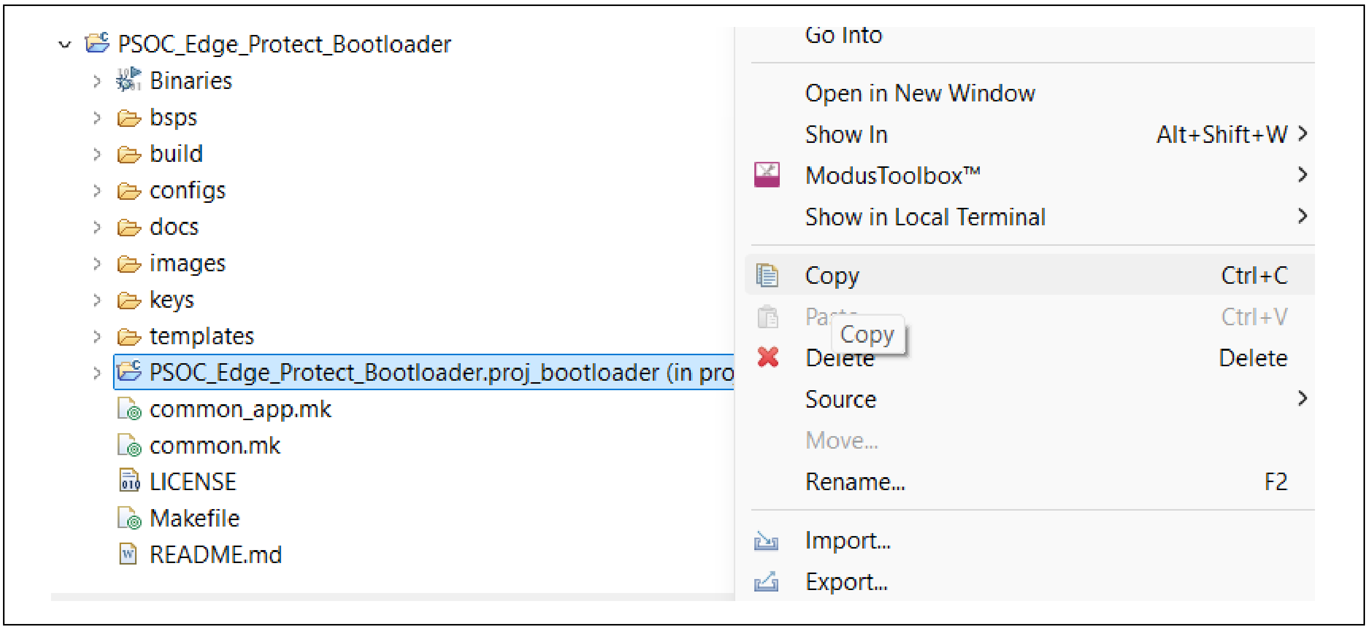


Figure 9 Copy proj_bootloader

10. Right click and paste the proj_bootloader to the root of Basic Secure App.

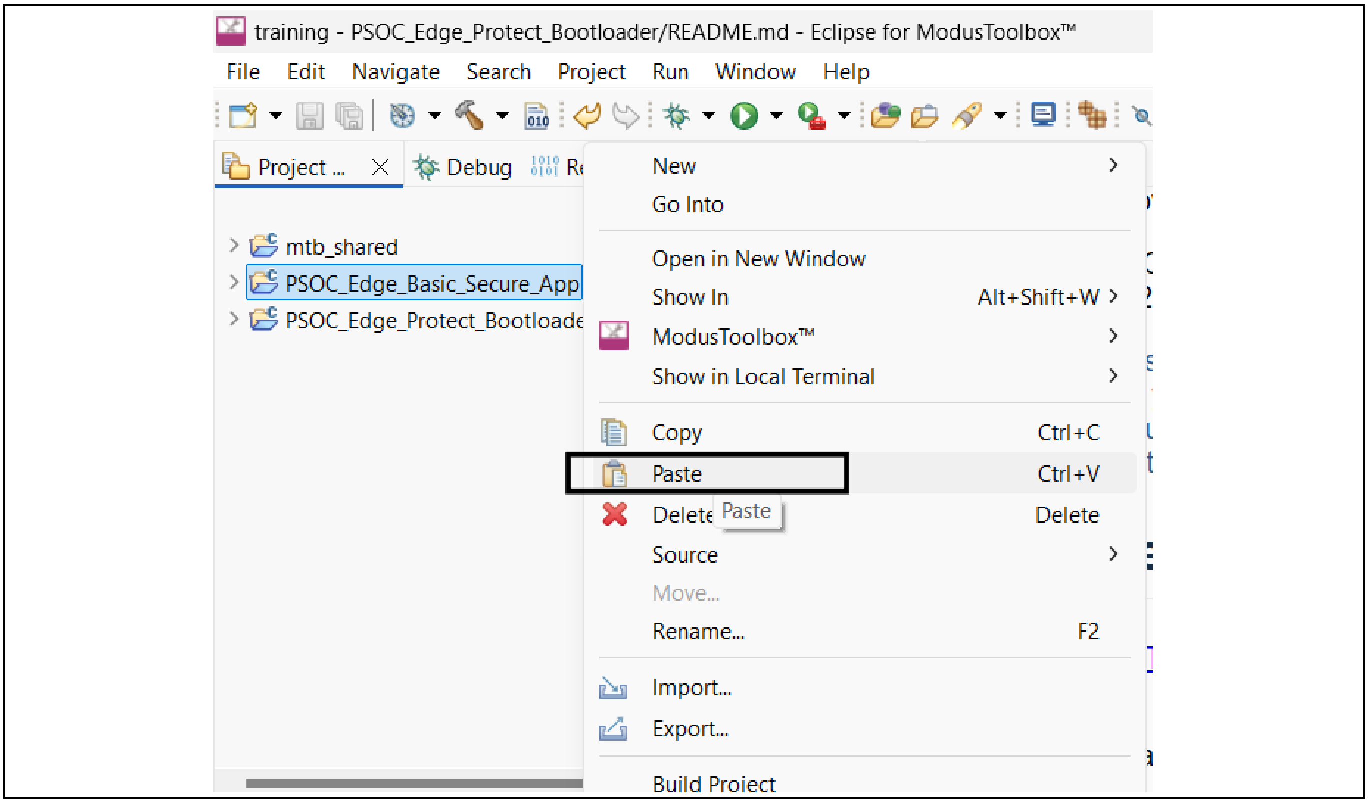


Figure 10 Paste proj_bootloader

11. While pasting edit the project name and location to point to the Basic Secure App.

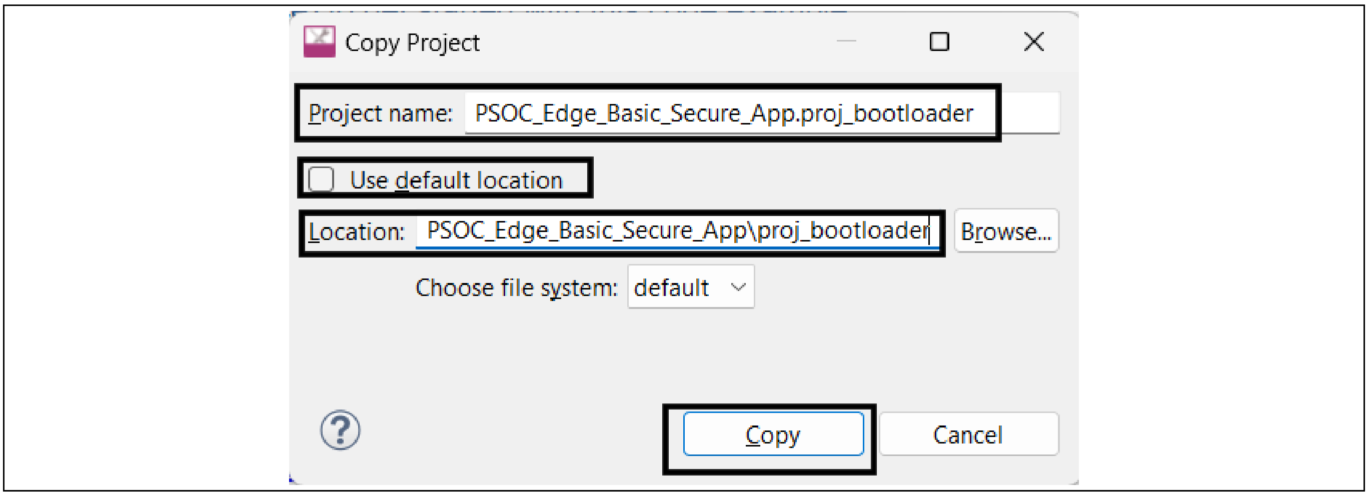


Figure 11 Edit the project name and location

12. Add the proj_bootloader to the list of projects in the Makefile of Basic Secure App.

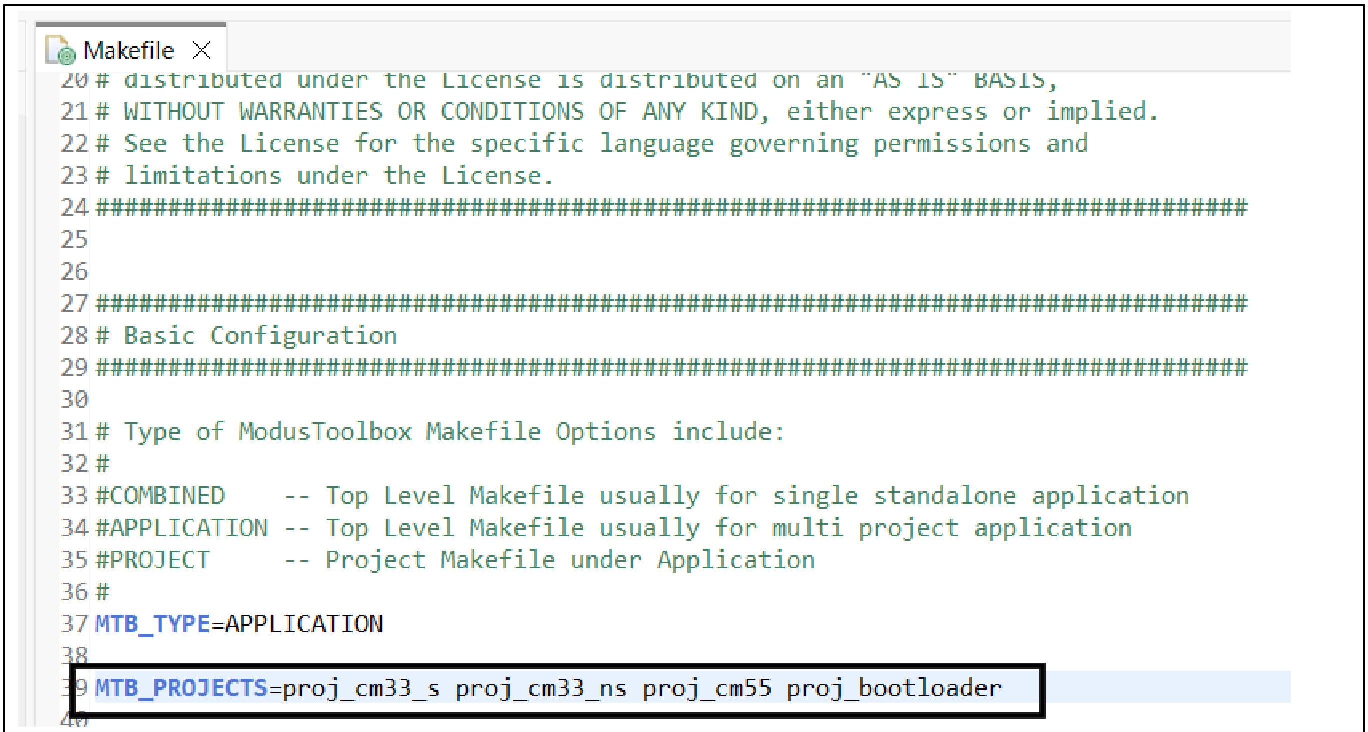


Figure 12 Add proj_bootloader to MTB projects list

13. Update the common.mk file of Basic Secure App to point to bootloader personality generated JSON file.

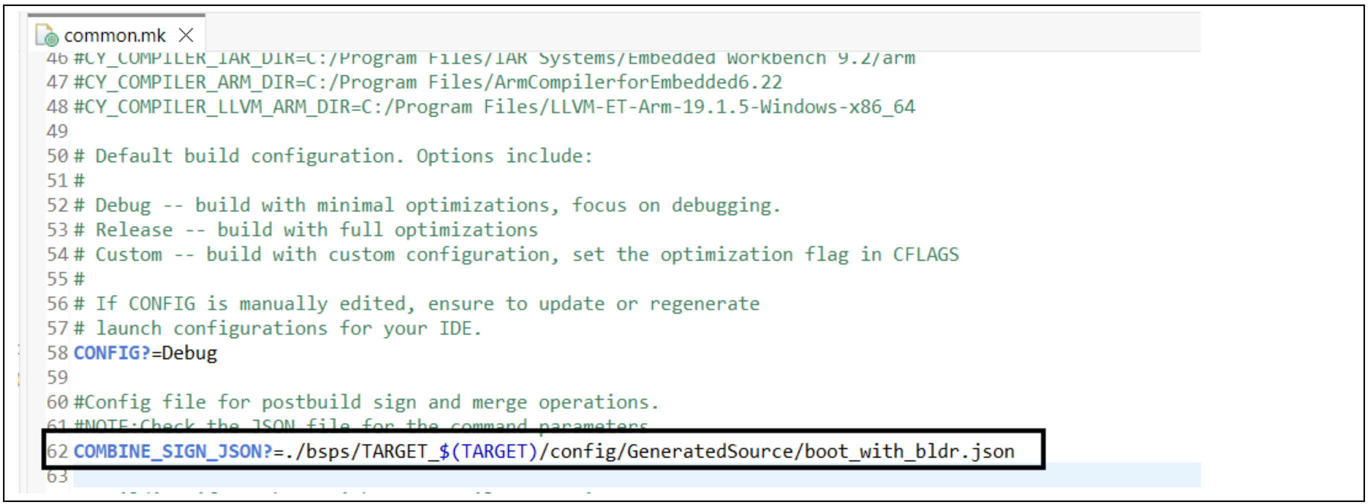


Figure 13 Update common.mk

14. Generate launches for Basic Secure App.

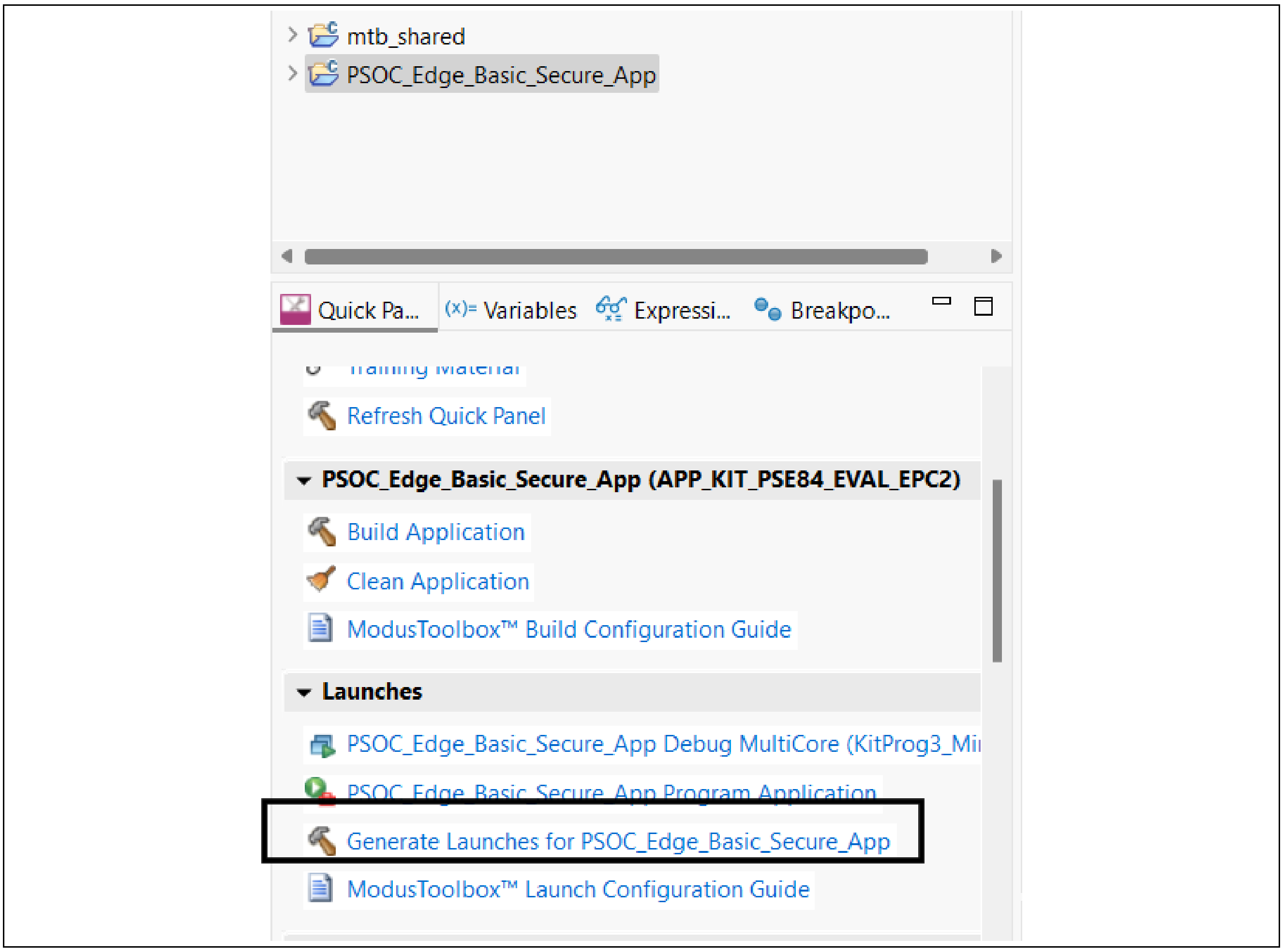


Figure 14 Generate launches for application

15. Now build and program the Basic Secure App by clicking the **Program Application** button in the Quick Panel.

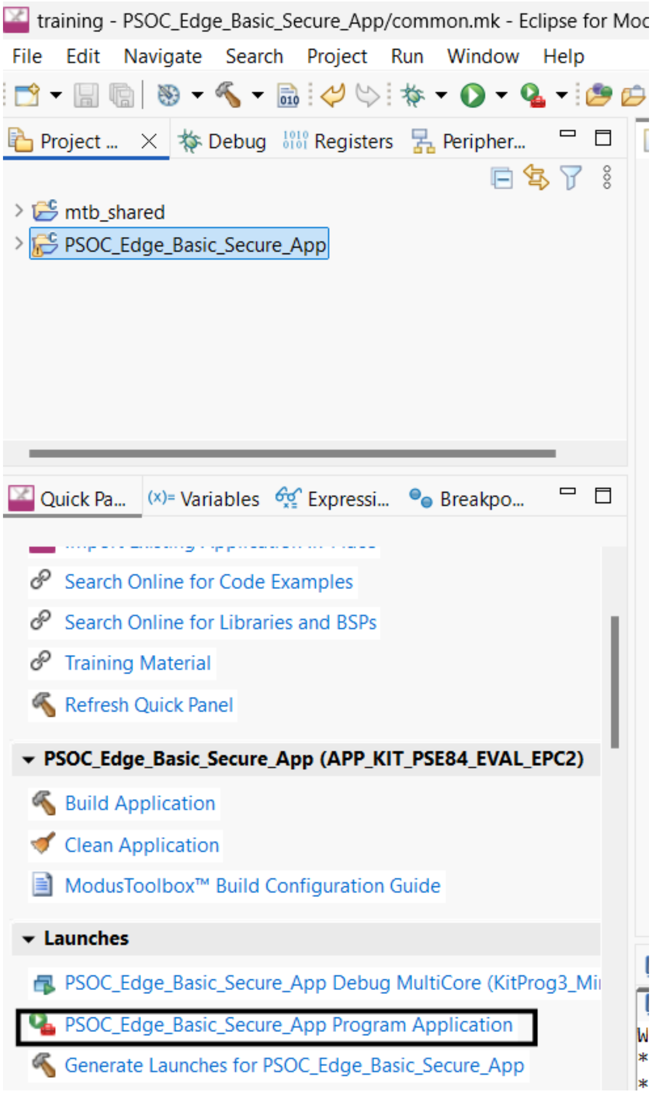


Figure 15 Program application

4.4 Basic Secure App booting to console

1. Set the Boot SW (SW6) / P17.6 position to "LOW / OFF". This will instruct Extended boot to boot the first application from RRAM. Edge Protect Bootloader is the first application in this case, which is built and programmed to RRAM in default case.
2. Reset / power cycle the device.
3. Observe the Tera Term console logs for Basic Secure App.

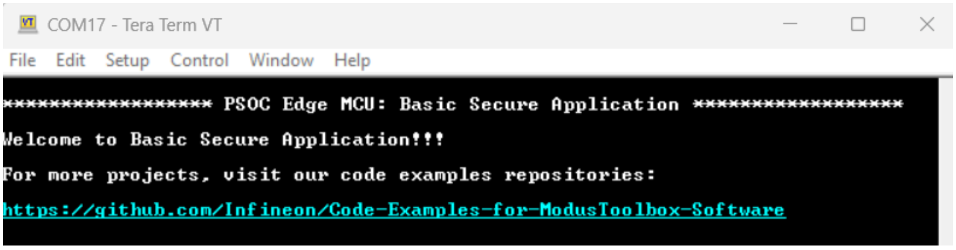


Figure 16 Terminal output for Basic Secure App

4. Observe the Tera Term console logs for Edge Protect Bootloader before the Basic Secure App logs.

Figure 17 Terminal output for bootloader

4.5 Conclusion

We successfully created and tested Edge Protect Bootloader for PSOC™ Edge E84. In the process we learnt:

- How to add proj_bootloader to Basic Secure Application.
- How to boot to a validated application using Edge Protect Bootloader.

5 Secure boot with Edge Protect Bootloader

5.1 Objective

Through this lab, you will gain a deeper understanding of secure boot, and the chain of trust involved, starting with the Infineon pre-programmed Extended Boot until the user application (Basic Secure App). You will learn how to sign bootloader and application images to extend the Chain of Trust (CoT).

5.2 Description

Enabling complete chain of trust requires Extended boot to verify Edge Protect Bootloader and Edge Protect Boot loader should verify the user application (all projects in user application). To do that, you will have to do the following:

1. Obtain the device ownership
2. Enable secure boot in OEM policy
3. Configure the bootloader solution personality to enable image validation
4. Build and program user application

5.3 Project Creation

Project creation process is same as discussed in section 4.3 above.

5.4 Enabling Secure boot

5.4.1 Obtain the device ownership

The process for transferring device ownership is detailed in the "Training Manual PSOC™ Edge E84 Getting Started with Security." Please adhere to the precise instructions outlined in the manual to generate the key pair and provision the device using edgeprotecttools to obtain device ownership. If you have previously obtained device ownership through this process, you can skip this step.

It is important to note that the same key pair (OEM_RoT_Key) used for provisioning must be used for signing and verification.

5.4.2 Enable Secure boot in policy

Secure boot is disabled by default. You can enable the Secure boot by modifying the default policy and provisioning it to the device. Secure boots must be enabled first in the policy, and new policy must be provisioned.

1. Edit <application-directory>/policy/policy_oem_provisioning.json file using any text editor to set the value field to true for the “secure_boot” parameter.

```
"extended_boot_policy": {
  "secure_boot": {
    "description": "Disable/Enable secure boot option",
    "value": true
  },
```

Figure 18 Extended boot policy

2. Connect the board to the host machine (PC) using a USB cable.

3. In modus-shell, navigate to <application-directory> and execute the following command to provision the device with the updated policy.

```
edgeprotecttools -t pse8xs2 provision-device -p policy/policy_oem_provisioning.json --key
keys/oem_private_key_0.pem
```

For further details, see the training manual at PSOC™ Edge E84 Training manual for Security Introduction.

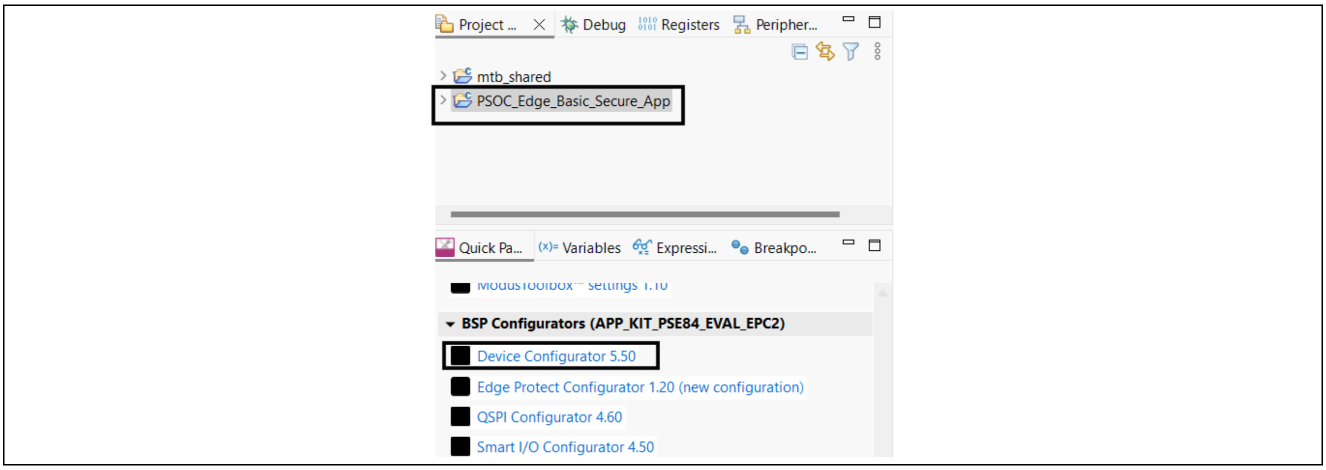
5.4.3 Configure the Edge Protect Bootloader for image validation

In the Enabling Secure boot section, we enabled Secure boot capability in Extended boot. The Extended boot will now validate the signature of the next image (Edge Protect Bootloader in this case) before booting it. You will have to sign the Edge Protect Bootloader image. Additionally you need to configure the Edge Protect Bootloader to validate the next images.

Follow steps below to enable EPB image validation and to sign the EPB image,

1. Ensure the oem_private_key_0.*pem* (pair of OEM_RoT_Key Public provisioned to the device) is available in the Basic_Secure_App/keys directory.

2. Open the device configurator for Basic Secure App.



3. Go the **Solutions** tab of the Device configurator, under the **Parameters** view of the **Edge Protect Bootloader solution** , enable the following:

   a) Validate boot.

   b) Validate upgrade.

   c) Enable bootloader signing (this option enables signing of bootloader image).

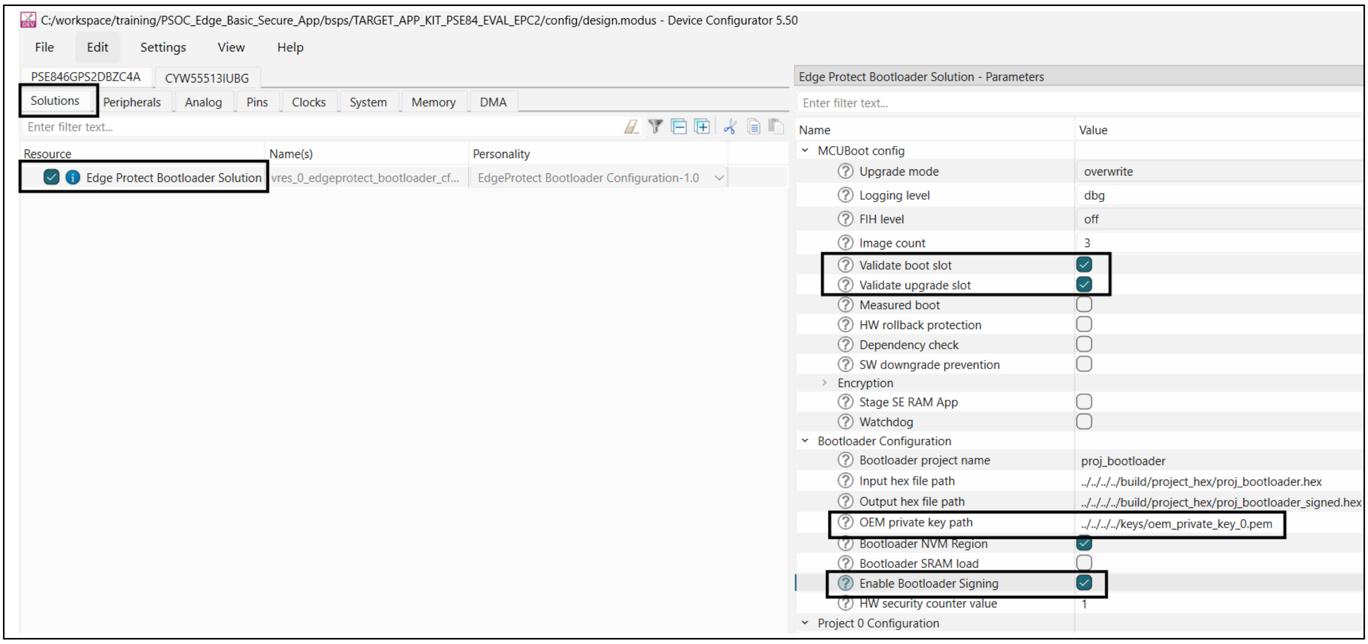


Figure 19 Enable image validation

4. Save your changes to the design.modus to generate bootloader configuration files.
5. Now build and program the Basic Secure App by clicking the **Program Application** button in the Quick Panel.

5.5 Conclusion

- We successfully enabled Secure boot of Edge Protect Bootloader and User application, thus established a chain of trust from Infineon programed software to user application.
- We learnt how to enable Secure boot capability at Extended boot and Edge protect Bootloader and how to sign each of these images.

Revision history

Document revision	Date	Description of changes
**	2025-11-13	Initial revision

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

Edition 2025-11-13

Published by

Infineon Technologies AG

81726 Munich, Germany

Do you have a question about this document?

Email:

erratum@infineon.com